Cognitive Stability Daemon (CSD): Stability Driven Access Control for LLMs

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Abstract

General purpose chatbots cannot guarantee life outcomes, nor can they negate free will or mental illness. Yet they *can* avoid amplifying instability. This paper proposes the **Cognitive Stability Daemon (CSD)**, a lightweight control plane that continuously estimates a user's *stability score* from conversation level signals (affective drift, coherence loss, burstiness, lexical risk), then gates access with a *cooldown* when risk exceeds thresholds. CSD changes safety from topic blocking to *user state awareness*. We formalize the stability vector, scoring, trip wire thresholds, back off policy, calibration, and privacy constraints; we situate the approach within digital phenotyping (Huckvale et al. 2019) and text based risk detection (Broadbent et al. 2023; Thomas and et al. 2025), and propose an evaluation protocol. The goal is to prevent *AI induced spirals* without paternalism, preserving agency while refusing to be an unsafe interlocutor.

Crisis support. If you are in immediate danger or thinking about harming yourself, call or text **988** in the U.S. for the Suicide & Crisis Lifeline (988 Suicide & Crisis Lifeline 2025). Follow reporting best practices (World Health Organization 2023).

1 Motivation

No safety filter can eliminate human unpredictability. However, interaction itself can modulate risk. Evidence from digital phenotyping shows passively measured behavioral signals can track clinical state (Huckvale et al. 2019). Separately, NLP can identify elevated suicide risk in text based counseling streams (Broadbent et al. 2023; Thomas and et al. 2025). We synthesize these literatures into *state aware access control*: when a user appears numerically unstable, the system pauses, de escalates, and resumes only after stability returns.

2 Related Work

Digital phenotyping. Smartphones and wearables enable moment to moment quantification of behavior (Huckvale et al. 2019). While we rely on *text only* signals to minimize intrusion, the framework is compatible with multimodal inputs.

Text based risk detection. Transformer models can flag self harm risk in crisis chats (Broadbent et al. 2023; Thomas and et al. 2025). Crisis Text Line reported 86% precision in identifying severe imminent risk early in conversations (Crisis Text Line 2018).

Affect models. We map language to valence arousal dominance (VAD) (Warriner et al. 2013) grounded in the circumplex model of affect (Posner et al. 2005).

Human dynamics. Bursty timing and heavy tails are characteristic of human communication (Barabási 2005); CSD treats temporal burstiness as a destabilization cue.

LLM safety. Constitutional AI and downstream moderation exemplify rules based harmlessness and output inspection (Bai et al. 2022; Sharma and et al. 2025). CSD complements these by conditioning on *user state*.

3 System Overview

CSD runs alongside the assistant in the control plane (Figure **??**, conceptual). On each user turn, CSD extracts features, updates a smoothed risk estimate, and either (i) permits normal generation, (ii) constrains the reply to fixed templates, or (iii) seals the session for a cooldown.

Key Design Goals

- **Stability first**: decide on user stability before generating content.
- **Privacy minimal**: compute from text only; store aggregates, not raw text.
- **Non paternalistic**: preserve agency; refuse to be an unsafe interlocutor.
- **Defense in depth**: combine state gating with output moderation.

4 Stability Vector and Score

Let x_t be a user turn at time t. We compute features f_t across linguistic, affective, lexical, coherence, and temporal families:

- **Affective drift:** valence and arousal estimates, sliding variance; LIWC style categories (Warriner et al. 2013).
- **Coherence drift:** 1 minus cosine similarity between consecutive turns.
- **Lexical risk:** catastrophizing markers; self harm intent score from a small on device classifier (binary probability).
- **Temporal burstiness:** coefficient of variation of inter message intervals; wordcount z scores; emphatic style (ALL CAPS, !!!).

Risk per turn is a calibrated weighted sum $r_t \in [0, 1]$, and a time decayed EWMA yields a smoothed risk s_t. Trigger when either $r_t \geq \tau_{\text{hard}}$ or $s_t \geq \tau_{\text{soft}}$, with an immediate hard gate for high self harm probability.

4.1 Reference Thresholds

Parameter	Default	Rationale
τ_{soft}	0.55	sustained elevation (3–5 turns)
τ_{hard}	0.80	spike guardrail
Self harm hard gate	0.65	high precision tripwire

5 Cooldown Policy

Upon trigger, CSD returns a low verbosity template (no improvisation), records a session token, and seals the interface for a back off interval:

$$\text{cooldown}(k) = \min\left(\max,\ \text{base} \cdot 2^{k-1}\right)$$

where k is consecutive strike count. Strikes decay when $s_t < \tau_{\text{recovery}}$.

6 Personalization and Calibration

CSD learns user baselines from prior stable sessions (means/variances of wordcount, burstiness, coherence drift) and caps personalization to avoid overfitting (e.g., weight deltas limited to $\pm 25\%$ of global). For n=1 case studies (e.g., heavy users), only aggregates are stored; raw text is discarded.

7 Governance, Privacy, Ethics

CSD is not diagnosis or treatment. It is *risk aware access control*. Documented policies should include: opt out for competent adults; guardian consent for minors; clear communication of what is measured; retention schedules; and independent audits. Pair with reporting best practices and crisis resources (World Health Organization 2023; 988 Suicide & Crisis Lifeline 2025).

8 Evaluation Protocol

1. **Offline:** simulate risk with public/self harm corpora; report ROC, PR, latency; measure false positives during benign chats.
2. **Online A/B:** compare (a) baseline moderation vs. (b) CSD + moderation; primary endpoints: reduction in unsafe assistant replies; secondary: user satisfaction after cooldown.
3. **Red team suites:** hopelessness spirals, fixation loops, rapid bursts, masked risk with neutral phrasing.
4. **Fairness:** subgroup performance; calibration drift monitoring.

9 Limitations

Text only inference can miss risk; some users will experience false positives. Cooldowns can be frustrating; therefore, templates must be compassionate and brief. Cultural and linguistic variance requires localized calibration (Huckvale et al. 2019). CSD should complement, not replace, human care.

10 Conclusion

You cannot outlaw free will or abolish mental illness. But a chatbot can refuse to be an amplifier. CSD reframes safety as state aware access control: *pause when unstable, resume when steady.* It is simple to implement, model agnostic, privacy minimal, and compatible with existing moderation.

Practical Appendix: Minimal Pseudocode

Per turn: extract features; compute r_t; update EWMA s_t; if hard gate or thresholds exceeded $\rightarrow$ lock; else generate; if near risk, reduce verbosity.

References

988 Suicide & Crisis Lifeline (2025). *Get Help Now*. Accessed Nov 9, 2025. URL: https://988lifeline.org/.

Bai, Yuntao et al. (2022). "Constitutional AI: Harmlessness from AI Feedback". In: *arXiv preprint arXiv:2212.08073*. URL: https://arxiv.org/abs/2212.08073.

Barabási, Albert-László (2005). "The origin of bursts and heavy tails in human dynamics". In: *Nature* 435.7039, pp. 207–211. DOI: 10.1038/nature03459.

Broadbent, Matthew et al. (2023). "A machine learning approach to identifying suicide risk among text-based crisis counseling encounters". In: *Frontiers in Psychiatry* 14, p. 1110527. DOI: 10.3389/fpsyt.2023.1110527.

Crisis Text Line (2018). *Detecting Crisis: An AI Solution*. Accessed Nov 9, 2025. URL: https://www.crisistextline.org/blog/2018/03/28/detecting-crisis-an-ai-solution/.

Huckvale, Kit, S. Venkatesh, and Helen Christensen (2019). "Toward clinical digital phenotyping: a timely opportunity to consider purpose, quality, and safety". In: *npj Digital Medicine* 2, p. 88. DOI: 10.1038/s41746-019-0166-1.

Posner, Jonathan, James A. Russell, and Bradley S. Peterson (2005). "The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology". In: *Development and Psychopathology* 17.3, pp. 715–734. DOI: 10.1017/S0954579405050340.

Sharma, R. and et al. (2025). *Early Stopping Harmful Outputs via Streaming Moderation*. arXiv preprint arXiv:2506.09996. URL: https://arxiv.org/abs/2506.09996.

Thomas, Jade and et al. (2025). "An Explainable AI Text Classifier for Suicidal and Self-Injurious Behaviors in a Real-World Crisis Helpline". In: *JMIR Public Health and Surveillance* 11.1, e63809. URL: https://publichealth.jmir.org/2025/1/e63809.

Warriner, Amy Beth, Victor Kuperman, and Marc Brysbaert (2013). "Norms of valence, arousal, and dominance for 13,915 English lemmas". In: *Behavior Research Methods* 45.4, pp. 1191–1207. DOI: 10.3758/s13428-012-0314-x.

World Health Organization (2023). *Responsible reporting on suicide: Quick reference guide*. Accessed Nov 9, 2025. URL: https://cdn.who.int/media/docs/default-source/mental-health/suicide/responsible-reporting-on-suicide.pdf.

License. Suggested: CC BY 4.0. **Companion code.** *CSD* reference implementation and test harness to be attached in repository.